TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Identification of Carbonyl Compounds

One reagent to catch every carbonyl, three gentle oxidants to split aldehydes from ketones, and the special tests that name formaldehyde, acetaldehyde, benzaldehyde and acetone.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 3 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to identify an unknown carbonyl compound — to prove the C=O group, decide whether the compound is an **aldehyde** or a **ketone**, and confirm individual members of the common set.

Aldehydes (R–CHO) and ketones (R–CO–R) share the carbonyl group, so the first test must be one that answers for both: 2,4-dinitrophenylhydrazine (Brady's reagent) condenses with any C=O and throws down a yellow-to-orange crystalline hydrazone —

$$R_2C{=}O + H_2N{-}NH{-}C_6H_3(NO_2)_2 \rightarrow R_2C{=}N{-}NH{-}C_6H_3(NO_2)_2\downarrow + H_2O$$

The split between the two classes is oxidation. An aldehyde's carbonyl carbon carries a hydrogen and is easily oxidised to the carboxylic acid, so aldehydes reduce mild oxidising agents that ketones cannot touch: Tollens' reagent (ammoniacal silver nitrate) deposits a silver mirror, Fehling's solution (alkaline Cu^{2+} held by tartrate) throws down red Cu_2O, and Schiff's reagent — a magenta dye bleached by SO_2 — has its colour restored. Ketones, with no aldehydic hydrogen, survive all three; forcing a ketone to oxidise needs vigorous conditions that break C–C bonds. Two border rules matter: Fehling's reagent is the weaker oxidant and fails with *aromatic* aldehydes such as benzaldehyde (Tollens still catches them), and the iodoform test cuts across the classification — it answers for the CH_3–CO– unit in methyl ketones *and* for acetaldehyde, the one aldehyde built on that unit.

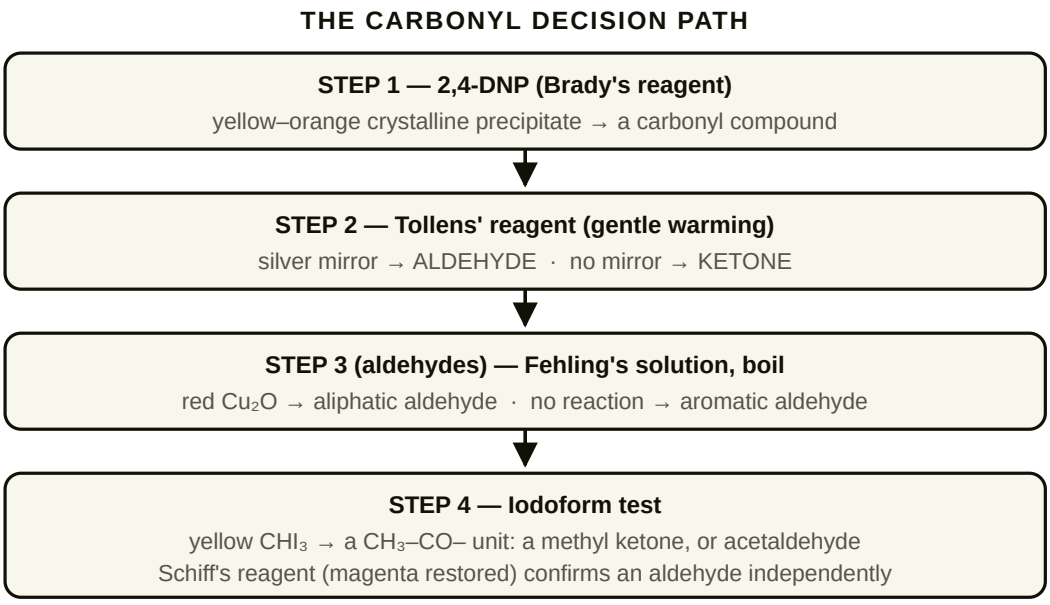


Figure 1 — The working order: one test for the whole class, one oxidant strong enough to split it, then the finer separations.

2. Apparatus and reagents

- Clean test tubes (one scrupulously clean tube for the mirror test), water bath, droppers, melting-point apparatus for a derivative if required; the unknown carbonyl compound.
- 2,4-dinitrophenylhydrazine reagent (Brady's reagent, in methanol/H_2SO_4 or ethanol/H_3PO_4), saturated sodium bisulphite solution, Tollens' reagent (prepared fresh: $AgNO_3$ + a drop of NaOH, then dilute NH_3 dropwise until the precipitate just dissolves), Fehling's A and B, Schiff's reagent, iodine solution and 10% NaOH (iodoform), sodium nitroprusside solution and NaOH (Legal's test for acetone).

3. Test scheme

TEST	ALDEHYDES	KETONES	CHEMISTRY
2,4-DNP	Yellow–orange crystalline ppt	Yellow–orange crystalline ppt	Condensation at C=O (equation in §1). The derivative's sharp melting point identifies the individual compound.
Sodium bisulphite	White crystalline addition compound	Only methyl ketones (and some cyclic ketones) react	Addition of HSO_3^- across C=O; the adduct is decomposed back to the carbonyl compound by

			dilute acid or alkali — also a purification method.
Tollens'	Silver mirror (or grey-black Ag)	No reaction	$RCHO + 2\ [Ag(NH_3)_2]OH \rightarrow RCOONH_4 + 2\ Ag\downarrow + 3\ NH_3 + H_2O$
Fehling's (boil)	Aliphatic: red Cu_2O ppt. Aromatic (benzaldehyde): **no reaction**	No reaction	$RCHO + 2\ Cu^{2+} + 5\ OH^- \rightarrow RCOO^- + Cu_2O\downarrow + 3\ H_2O$. Fehling's is weaker than Tollens' — the aromatic failure is the classic trap.
Schiff's	Magenta/pink colour restored within minutes	Remains colourless	The aldehyde restores the quinonoid dye structure that SO_2 had bleached.
Iodoform	Only acetaldehyde positive	Methyl ketones positive	$CH_3COCH_3 + 3\ I_2 + 4\ NaOH \rightarrow CHI_3\downarrow + CH_3COONa + 3\ NaI + 3\ H_2O$

Confirming individuals. Formaldehyde (formalin): pungent, choking vapour; Schiff's colour returns at once and strongly; Tollens mirror forms very readily — formaldehyde is the strongest reducing aldehyde here (it can be oxidised twice, to formic acid and then CO_2). **Acetaldehyde:** the only aldehyde positive to iodoform; Fehling and Schiff both positive. **Benzaldehyde:** oil-of-bitter-almonds odour; Tollens positive but Fehling **negative**; with concentrated KOH it undergoes the Cannizzaro reaction — $2\ C_6H_5CHO + KOH \rightarrow C_6H_5CH_2OH + C_6H_5COOK$ — one molecule oxidised, one reduced, possible only because it has no α-hydrogen. **Acetone:** sweet solvent odour, fully miscible with water; iodoform positive; Legal's test — sodium nitroprusside then NaOH — gives a red colour; Tollens, Fehling and Schiff all negative, as a ketone should be. **Acetophenone:** iodoform positive (a methyl ketone), sweet odour, no reducing tests.

4. Summary decision table

COMPOUND	TOLLENS	FEHLING	SCHIFF	IODOFORM	DECISIVE EXTRA

Formaldehyde	+ (ready mirror)	+	+ (immediate)	−	Pungent gas; strongest reducer of the set
Acetaldehyde	+	+	+	+	Only aldehyde answering iodoform
Benzaldehyde	+	−	+	−	Bitter-almond odour; Cannizzaro with conc. KOH
Acetone	−	−	−	+	Legal's test: red colour with nitroprusside + NaOH
Acetophenone	−	−	−	+	Methyl ketone; sweet odour; 2,4-DNP derivative
Other ketones (e.g. butan-2-one is a methyl ketone; diethyl ketone is not)	−	−	−	Only if CH_3–CO–	2,4-DNP ppt + derivative m.p.

Every row gives the 2,4-DNP precipitate — read the table only after Step 1 of Figure 1 has admitted the unknown to the carbonyl class.

5. Model identification record

WORKED EXAMPLE — UNKNOWN C

Colourless liquid with a sweet, solvent-like odour; miscible with water.

2,4-DNP: orange crystalline precipitate. → A **carbonyl compound**.

Tollens: no mirror. Fehling (boiled): no red precipitate. Schiff: stays colourless. → A **ketone**, three times over.

Iodoform: yellow precipitate of CHI_3. → A **methyl ketone**, CH_3–CO–.

Legal's test: red colour with sodium nitroprusside and NaOH. → Confirms acetone among the methyl ketones.

Conclusion: unknown C is acetone (propanone).

6. Precautions

- **Tollens' reagent: prepare fresh, use at once, destroy the residue.** On standing, ammoniacal silver solutions deposit explosive "fulminating silver" (silver nitride). After the test, acidify all residues with dilute HNO_3 and wash them away — never store the reagent, never let it dry in the tube.
- The mirror test needs a chemically clean tube (wash with warm NaOH, then distilled water): grease gives a patchy grey deposit instead of a mirror, which is still a positive — read it as silver, not as failure.
- Warm Tollens and Fehling mixtures in a water bath, never over a naked flame; several unknowns are flammable solvents.
- 2,4-DNP reagent contains strong acid and is itself a suspect carcinogen — gloves, no skin contact, and keep the dry solid away from friction and flame (it is stored damp for a reason).
- Schiff's test: judge within a few minutes and against a blank — very slow colour drift in air is not a positive.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Benzaldehyde gives a silver mirror with Tollens' reagent but does not reduce Fehling's solution. Why the difference?
2. Why must Tollens' reagent be freshly prepared, and how is it disposed of?
3. An unknown gives an orange 2,4-DNP precipitate and a yellow iodoform precipitate, but no silver mirror. What class and what structural unit does it contain?
4. Of formaldehyde and acetaldehyde, which gives the iodoform test, and why?
5. What is Schiff's reagent, and what does an aldehyde do to it?
6. In the iodoform test on acetone, what mass of iodoform is formed from 5.8 g of acetone, assuming complete reaction?

Answer key

1. The two reagents are oxidants of different strength. Tollens' reagent (Ag^+ in ammonia) is strong and sensitive enough to oxidise benzaldehyde to benzoic acid. Fehling's Cu^{2+}, held in alkaline tartrate, is a milder oxidant that oxidises aliphatic aldehydes but not the resonance-stabilised aromatic aldehyde under the test conditions.
2. On standing, the ammoniacal silver solution deposits silver nitride ("fulminating silver"), which is violently explosive when dry. Fresh preparation avoids its formation; residues are destroyed at once with dilute nitric acid, which dissolves any deposit safely.

3. A ketone containing a CH_3–CO– (methyl ketone) unit: carbonyl proven by 2,4-DNP, ketone proven by the absent mirror, and the iodoform demands the acetyl unit — acetone or acetophenone, not a general ketone.
4. Acetaldehyde. It is CH_3–CHO — it carries the CH_3–C(=O)– unit the iodoform reaction cleaves to CHI_3. Formaldehyde, HCHO, has no CH_3 group at all and cannot form iodoform.
5. Schiff's reagent is the magenta dye rosaniline (basic fuchsin) decolourised by sulphur dioxide. An aldehyde combines with the bleached dye and restores its coloured quinonoid structure, so the pink-magenta colour returns; ketones do not restore it.
6. 5.8 g acetone = 5.8/58 = 0.10 mol. The equation gives 1 mol CHI_3 per mol acetone, so CHI_3 formed = 0.10 mol × 393.7 g mol^{-1} = **39.4 g**.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib